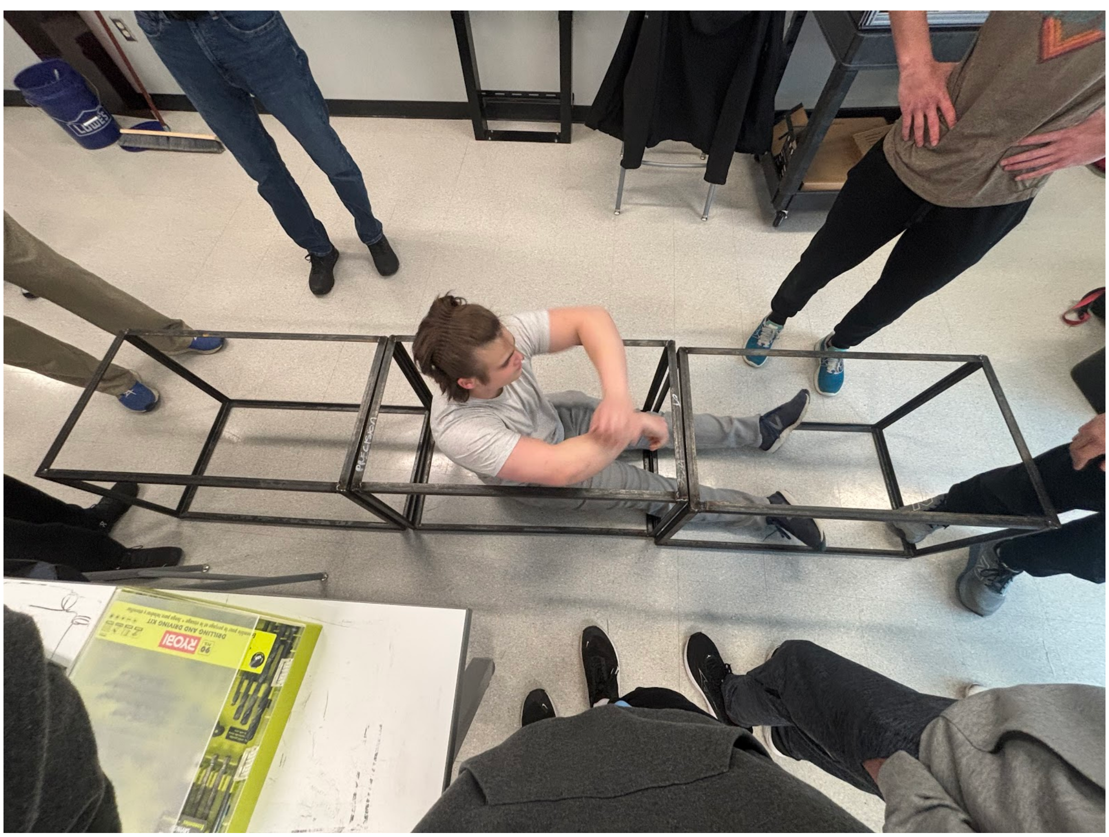
Prototype testing and data collection plan

Testing Plan:

The testing plan is designed to address the key design requirements of the Komodo dragon containment system, with an emphasis on the primary functions and structural integrity of the system. The plan outlines the steps for evaluating the system's performance in areas like strength, usability, adjustability, and long-term durability.

Frame and Guillotine Door Testing:

The guillotine door provides controlled access for zookeepers while ensuring the safety of the Komodo dragons. Our testing focused on the door's sliding mechanism and the effectiveness of the vertical movement and its ease of operation. We assessed the door's friction levels to ensure it moved smoothly within its frame by sliding it up and down and feeling how easily it moved. Performance was measured by the door's ability to open and close without resistance, to achieve an optimal sliding action for the zookeepers' use.



Box Connection System Testing:

The box connection system plays a critical role in maintaining the integrity of the containment system. Testing will primarily focus on evaluating the strength and reliability of the connection mechanism. A key part of the plan involves applying force to the connection points to assess their resistance to stress and to determine whether the system can withstand the weight and force generated during use. Additionally, the adjustability of the connection system will be tested to ensure that the boxes can be reconfigured easily and securely as needed.

Mathematical modeling was used to determine the shear stress and shear force that the box connection system can withstand. The modeling from Element F is given below:

Given the equation for shear stress $\tau = \frac{dV}{dA}$, we may approximate the stress on a particular object if we take the shear force (V) to be constant. This provides the following derivation:

$$\tau = \frac{dV}{dA}$$

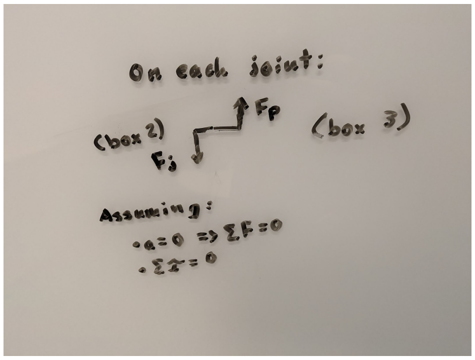
$$\tau dA = dV$$

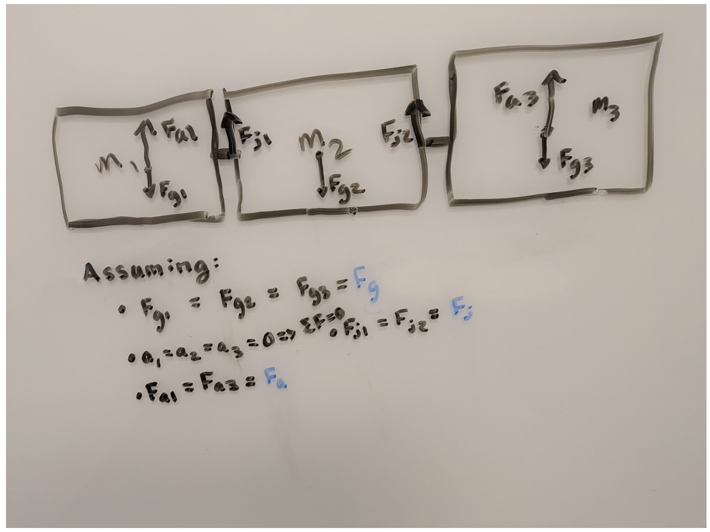
$$\int \tau dA = \int dV$$

$$\tau_{avg} \int dA = V$$

$$\tau_{avg} = \frac{V}{A}$$

Following this equation, we now know that we must find the shear force acting on the latch (V) and the cross-sectional area upon which the force acts. To begin, we will determine the force that acts on the connection system between two boxes, drawing a free body diagram first to make the subsequent derivations more intuitive.





We can now further analyze only the middle box under static conditions to determine what the shear force is on the joint:

$$\sum F_2 = m_2 a = |F_{j1}| + |F_{j2}| - |F_g|$$

$$m_2(0) = 2|F_j| - |F_g|$$

$$0 = 2|F_j| - m_2 g$$

$$|F_j| = \frac{m_2 g}{2}$$

Thus we know $V = \frac{m_2 g}{2}$, and we are now left only to determine the cross-sectional area of the joint that connects the boxes. Since the latch that we are discussing has two parallel bars joined at the end with minimal slack between the two (a similar version is shown below), only one of the bars will be in contact with the plate they attach to when a load is applied perpendicular to the bars on the plane that connects the two bars. That means that $A = n\pi r^2 = 0.000004\pi n$, where n is the number of latches. Substituting this into the equation from earlier, we find $\tau_{avg} = \frac{m_2 g}{0.000008\pi n^2}$

Now to find the angle of deflection we can expect in the system, we can use Hooke's Law for shear: $\gamma = \frac{\tau}{G}$, where $\gamma = \frac{\Delta x}{L} = tan(\theta)$ and G is the shear modulus (79.3 GPa for steel), meaning we can rewrite this equation as follows to solve for the angle of displacement:

$$\gamma = tan(\theta) = \frac{\tau}{G}$$

$$\theta = tan^{-1}(\frac{\tau}{(79.3)(10^9)})$$

$$\theta = tan^{-1}(\frac{m_2 g}{634400\pi n^2})$$

Now, to solve for the number of latches needed to keep the angle of displacement under 5 degrees (arbitrarily chosen for example although we will reduce this to 0.001 degrees later), we can rewrite the equation as follows:

$$tan(5 * \frac{\pi}{180}) = \frac{m_2 g}{634400\pi n^2}$$

$$634400\pi n^2 = \frac{m_2 g}{tan(5*\frac{\pi}{180})}$$

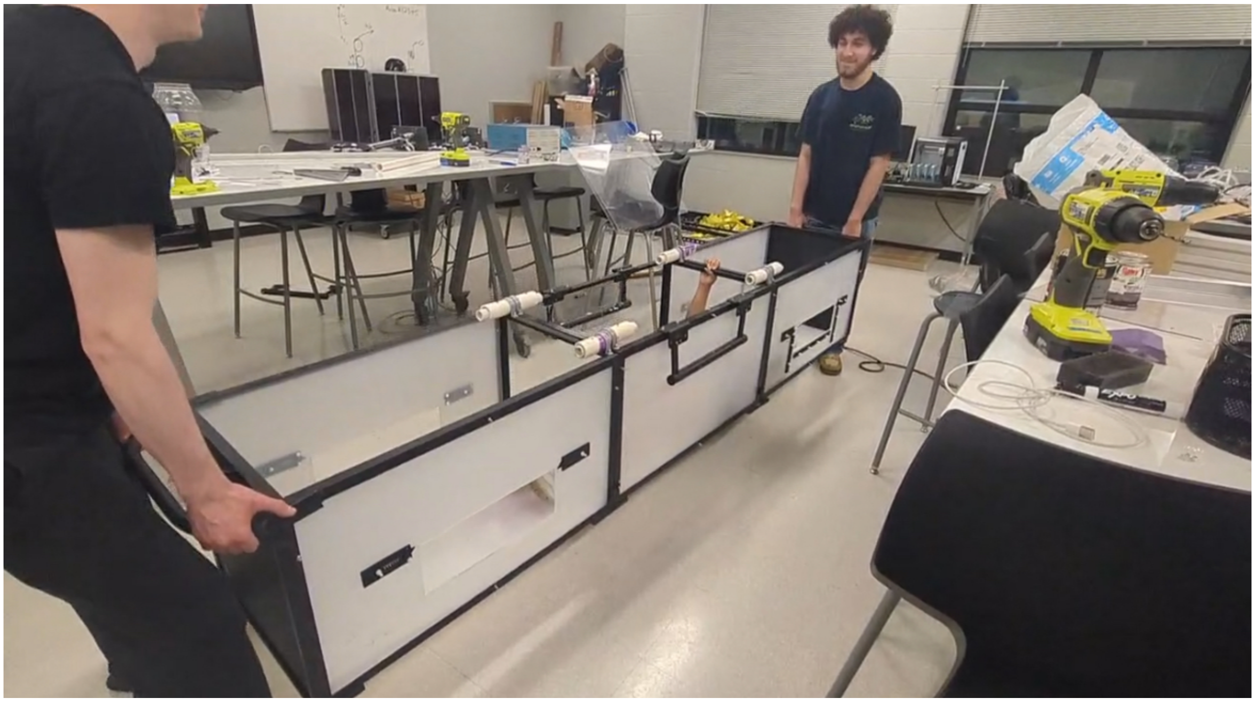
$$n = \sqrt{\frac{m_2 g}{(634400)(\pi)tan((5)(\frac{\pi}{180}))}}$$

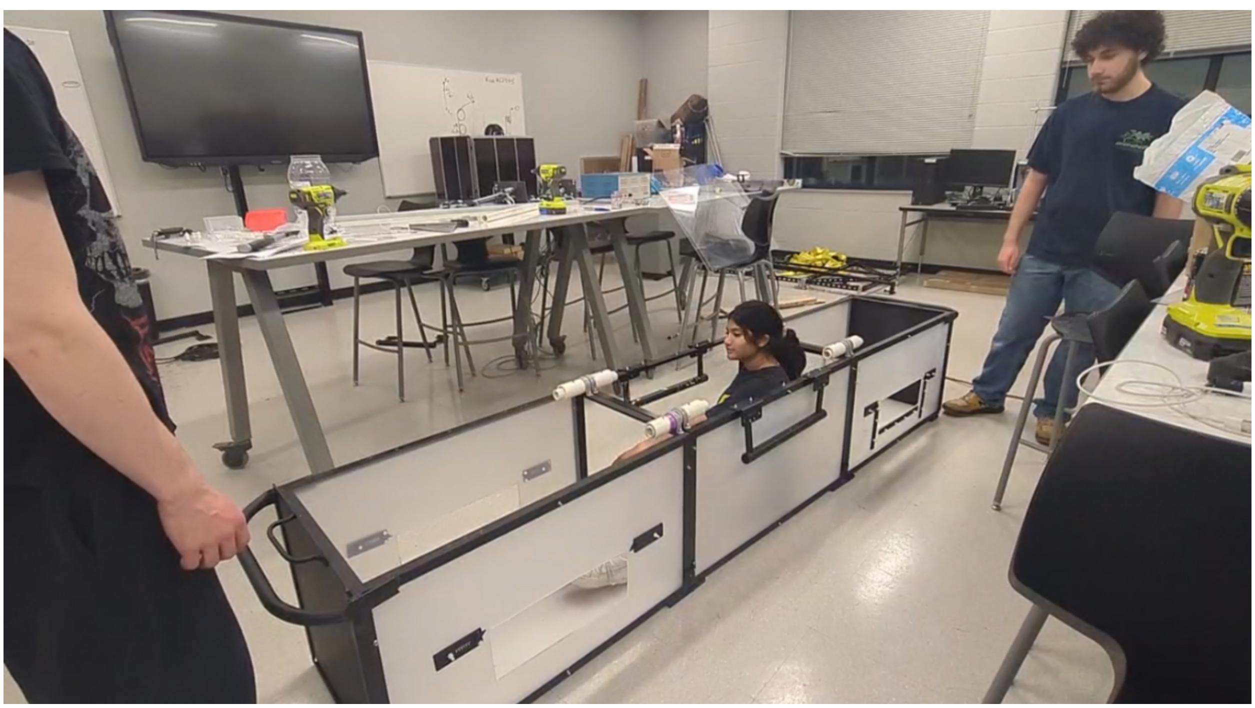
Substituting known values leaves us with:

$$n = \sqrt{\frac{(33.3)(9.81)}{(634400)(\pi)tan((5)(\frac{\pi}{180}))}} \approx 0.0432 \text{ Latches}$$

The Box connection system was also tested through a series of trials, where a classmate was placed inside the containment system with the completed connection system and lifted up and held for a period of time. The box was moved around without issue, and could be shaken without disconnecting.

Mr. Kiser noted a system of testing the box that including putting weights inside the box while increasing until we exceed a set value as a certain percentage (e.g. 20%) over the max possible weight would be suitable. When testing, we were able to determine that the box could hold approximately 250 lbs without failure, which is over the weight of the dragon by a decent margin and thus tests the performance of the box objectively.





Data Collection and Performance Metrics:

The testing plan also includes comprehensive data collection methods, which will help quantify the effectiveness of each design component. For the frame and guillotine door, performance metrics will include ease of movement, friction levels, and the door’s ability to stay securely in place. In terms of ease of movement, the doors could be easily assembled and disassembled, requiring 1.5 seconds for disassembly and 3 seconds for assembly. The coefficient of friction was minimized with the primer, which makes it harder for the doors to be opened while being pushed against. To test the ability of the door to stay in place, the box was held upside down, and the door remained as it was. The box connection system will be tested for tensile strength, with data gathered on the force it can withstand before failure, but there are no extra clamps currently, so we have resorted to purely modeling as of now.

Limitations and Future Considerations:

While the testing plan covers essential aspects of the design, some limitations exist due to factors that are difficult to test in a prototype form. For instance, long-term durability of the box connection system may not be fully assessable in the short-term testing, especially when it comes to repeated disassembly and reassembly. The long-term frictional behavior of the guillotine door under prolonged use may require extended testing periods to fully understand. These aspects will be reviewed in greater detail after further refinement of the system.